

WEEK-7

Task-1: To parse Input Redirection, Open Files, Redirect stdin, Handle Missing Files, Restore Streams, Test File Inputs

CODE:

```
pavan@DESKTOP-580LEIN: ~/input_project
GNU nano 8.7.1 input.c *
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main() {
    int fd;
    char buffer[100];

    fd = open("input.txt", O_RDONLY);

    if (fd == -1) {
        perror("Error opening input.txt");
        return 1;
    }

    dup2(fd, STDIN_FILENO);
    close(fd);

    printf("Reading from input.txt:\n");

    while (fgets(buffer, sizeof(buffer), stdin))
        printf("%s", buffer);

    return 0;
}
```

OUTPUT:

```
pavani@DESKTOP-380LEIN:~/input_project$ gcc input.c -o input
pavani@DESKTOP-380LEIN:~/input_project$ ./input
Reading from input.txt:
Hello
This is input redirection.
Ubuntu Linux.
pavani@DESKTOP-380LEIN:~/input_project$ _
```

Report: The input redirection program was successfully implemented in Ubuntu.

A file was opened using open() and connected to standard input using dup2().

The program successfully read and displayed the contents of the input file.

Missing files were handled using an appropriate error message.

Task-2: To Parse Output Redirection, Create Output Files, Redirect stdout, Handle Permissions, Verify Output Data, Test Redirection

Code:

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main() {
    int fd;

    fd = open("output.txt",
              O_WRONLY | O_CREAT | O_TRUNC,
              0644);

    if (fd == -1) {
        perror("Error opening output.txt");
        return 1;
    }

    dup2(fd, STDOUT_FILENO);
    close(fd);

    printf("Hello from output redirection!\n");
    printf("This data is stored in a file.\n");

    return 0;
}_
```

Output:

```
pavani@DESKTOP-380LEIN:~$ mkdir output_project
pavani@DESKTOP-380LEIN:~$ cd output_project
pavani@DESKTOP-380LEIN:~/output_project$
pavani@DESKTOP-380LEIN:~/output_project$ nano output.c
pavani@DESKTOP-380LEIN:~/output_project$ gcc output.c -o output
pavani@DESKTOP-380LEIN:~/output_project$ ./output
pavani@DESKTOP-380LEIN:~/output_project$ output.txt
output.txt: command not found
pavani@DESKTOP-380LEIN:~/output_project$ cat output.txt
Hello from output redirection!
This data is stored in a file.
pavani@DESKTOP-380LEIN:~/output_project$ _
```

Report: The output redirection program was successfully implemented in Ubuntu. Standard output was redirected from the terminal to an output file using `dup2()`. The program created the file and stored the generated output inside it. The output data was verified using the `cat` command.

Task-3: To Parse Append Operator, Open Files in Append Mode, Redirect Streams, Preserve Existing Data, Validate Appends, Test Scenarios

Code:

```

#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main() {
    int fd;

    fd = open("data.txt",
              O_WRONLY | O_CREAT | O_APPEND,
              0644);

    if (fd == -1) {
        perror("Error opening data.txt");
        return 1;
    }

    dup2(fd, STDOUT_FILENO);
    close(fd);

    printf("New line added.\n");

    return 0;
}

```

Output:

```

pavani@DESKTOP-380LEIN: ~/skill
pavani@DESKTOP-380LEIN:~$ mkdir skill
pavani@DESKTOP-380LEIN:~$ cd skill
pavani@DESKTOP-380LEIN:~/skill$ nano append.c
pavani@DESKTOP-380LEIN:~/skill$ gcc append.c -o append
pavani@DESKTOP-380LEIN:~/skill$ ./append
pavani@DESKTOP-380LEIN:~/skill$ cat data.txt
New line added.
pavani@DESKTOP-380LEIN:~/skill$

```

Report: The append redirection program was successfully implemented in Ubuntu. The file was opened using O_APPEND so that new data was added at the end.

Existing file contents were preserved during multiple executions.
The appended data was verified successfully using the cat command.

Task-4: TO parse stderr Redirection, Redirect Error Streams, Capture Error Output, Verify File Contents, Handle Failures, Test Error Cases

Code:

```
GNU nano 8.7.1 stderr.c *
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main() {
    int fd;

    fd = open("error.txt",
              O_WRONLY | O_CREAT | O_TRUNC,
              0644);

    if (fd == -1)
        return 1;

    dup2(fd, STDERR_FILENO);
    close(fd);

    FILE *file = fopen("missing.txt", "r");

    if (file == NULL) {
        perror("Error");
        return 1;
    }

    fclose(file);

    return 0;
}
```

Output:

```
pavani@DESKTOP-380LEIN:~/skill$ nano stderr.c
pavani@DESKTOP-380LEIN:~/skill$ gcc stderr.c -o stderr
pavani@DESKTOP-380LEIN:~/skill$ ./stderr
pavani@DESKTOP-380LEIN:~/skill$
pavani@DESKTOP-380LEIN:~/skill$
pavani@DESKTOP-380LEIN:~/skill$ cat error.txt
Error: No such file or directory
```

Report: The stderr redirection program was successfully implemented in Ubuntu.

The standard error stream was redirected to a file using dup2().

An intentional file-opening failure was generated to test error handling.

The error message was successfully captured and verified inside error.txt.